A D PATEL INSTITUTE OF TECHNOLOGY (ADIT)
DEPARTMENT OF INFORMATION TECHNOLOGY
CVM UNIVERSITY

MINI PROJECT REPORT

SpendWise - Expense & Budget Tracker

An Offline Personal Financial Management Mobile Application

Subject	Mobile Application Development (MAD)
Submitted by	Heel Soni (Solo Developer)
Enrollment No.	12302080601034
Department / Class	Information Technology, IT-A (B8A)
Semester	VII
Academic Year	2026 – 2027
Technologies Used	Flutter, Dart, SQLite (sqflite), fl_chart, intl, Git

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Faculty Signature Date

Certificate of Approval

This is to certify that the mini-project report entitled "SpendWise - Expense & Budget Tracker" has been successfully completed by Heel Soni (Enrollment No.: 12302080601034), student of Seventh Semester, Department of Information Technology, A D Patel Institute of Technology, CVM University, in partial fulfillment of the academic requirements for the subject Mobile Application Development (MAD) during the academic year 2026–2027.

The work presented herein represents an authentic and independent project developed under regular faculty guidance.

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Internal Guide / Faculty	Head of Department (IT)
Department of Information Technology	Department of Information Technology
ADIT, CVM University	ADIT, CVM University

Candidate Declaration

I, Heel Soni (Enrollment No.: 12302080601034), hereby declare that the project entitled "SpendWise - Expense & Budget Tracker" is my own original work developed as a solo project. All architectural concepts, database designs, Dart/Flutter implementations, and test verifications have been performed by me. This project has not been submitted previously to any other university or examining body for any degree or diploma.

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Abstract

Effective personal financial management is critical for individuals and college students navigating constrained monthly budgets. While modern application stores feature numerous expense management tools, the vast majority mandate persistent internet connectivity, third-party user authentication, invasive cloud tracking, or paid premium subscriptions.

To overcome these barriers, SpendWise was developed as a clean, responsive, and completely offline personal expense and budget management mobile application built with Flutter (v3.47+ Stable), Dart, and SQLite (sqflite). SpendWise operates strictly on-device with zero reliance on cloud APIs, user sign-ups, or remote servers, guaranteeing total data sovereignty and privacy.

The system empowers users with full CRUD (Create, Read, Update, Delete) transaction management, monthly budget tracking, real-time remaining balance calculations, automated over-budget threshold visual alerts, and interactive categorical distribution charts rendered via fl_chart. The database architecture follows the Singleton Pattern with SQLite ACID-compliant storage and ISO 8601 string-based date querying. Comprehensive testing comprising 10 functional and unit test cases validated 100% input integrity, reliable serialization, and zero computational errors.

Keywords: Mobile Application Development, Flutter, Dart, SQLite, sqflite, Personal Finance, Expense Tracking, Budget Management, fl_chart, Material 3, Offline Architecture.

1. Introduction

1.1 Background & Motivation

In an increasingly digitized economy characterized by contactless payments, credit facilities, and recurring subscriptions, keeping track of daily micro-expenditures is challenging. For college students living on predetermined monthly allowances, unintentional overspending frequently leads to financial distress before the end of the calendar month.

Manual bookkeeping on physical paper or spreadsheets is tedious, error-prone, and rarely consulted on-the-go. SpendWise delivers an offline, frictionless, and instantaneous recording companion directly in the student's pocket.

1.2 The Importance of Personal Budgeting

Budgeting empowers individuals to measure real spending against planned financial limits. Knowing exact monthly totals, remaining balances, and categorical distributions (Food, Travel, Bills, etc.) encourages proactive discipline rather than reactive panic.

1.3 Why a Native Mobile Solution?

Transactions happen everywhere—in cafeterias, buses, grocery stores, and bookstores. A native Flutter mobile app delivers smooth 60+ FPS performance, instant cold start under 1 second, and complete offline autonomy via embedded SQLite storage.

2. Problem Statement

2.1 Existing System Deficiencies

1. Mandatory Cloud Connectivity: Most commercial apps fail completely when mobile data is absent.
2. Privacy Infringements: Financial transactions and spending habits are uploaded to remote servers.
3. Authentication Overkill: Requiring emails, passwords, and OTPs creates high friction for logging minor ₹30 snacks.
4. Intrusive Advertisements: Ad banners interrupt users and increase battery and bandwidth usage.
5. Lack of Dynamic Warnings: Passive notebooks don't alert users when spending exceeds allocated limits.

3. Objectives

3.1 Key Goals

- Architect a 100% offline ACID-compliant SQLite relational database using sqflite.
- Implement full CRUD transaction management with swipe-to-delete confirmation and Undo.
- Provide monthly budget tracking with dynamic remaining balance calculations.
- Implement automated over-budget warning banners with visual color indicators.
- Integrate interactive pie charts using fl_chart for category analytics.

• Offer multi-category horizontal chip filters and month navigation selectors.
• Guarantee strict client-side validation for amounts, titles, and dates.

4. Literature & Existing System Study

4.1 Comparison with Existing Systems

The table below compares SpendWise against traditional manual diaries, spreadsheets, and commercial cloud applications:

Feature	Paper Diary	Google Sheets	Cloud Apps	SpendWise
Offline Operation	Full	Limited	No / Degraded	**100% Native Offline**
Privacy & Security	Physical Risk	Cloud Stored	Third-party Monitored	**100% Local Device Storage**
Launch Speed	High	Low	Low (Login/Sync)	**Instant (< 1s)**
Interactive Charts	None	Manual Setup	Paywalled	**Real-time fl_chart**
Over-Budget Alert	None	None	Delayed	**Instant Visual Banner**

5. System Architecture & Technology Stack

SpendWise employs a clean layered architecture consisting of Presentation, Domain & Validation, Data Access, and Local Storage layers:

1. Presentation Layer: Built with Flutter Material 3 widgets (HomeScreen, AddEditScreen, SummaryScreen, ExpenseTile).
2. Domain Layer: Encapsulates Expense data models and pure validation logic (Validators).
3. Data Access Layer: Singleton DBHelper ensuring unified thread-safe SQLite operations.
4. Storage Layer: Embedded spendwise.db SQLite database featuring 'expenses' and 'budgets' tables.

6. Database Design

Table: expenses (SQLite Table Schema)

Column	Data Type	Constraint	Description

id	INTEGER	PRIMARY KEY AUTOINCREMENT	Unique expense identifier
title	TEXT	NOT NULL	Expense description (e.g., Lunch)
amount	REAL	NOT NULL	Cost spent in INR (> 0)
category	TEXT	NOT NULL	Category (Food, Travel, Bills, etc.)
date	TEXT	NOT NULL	ISO 8601 timestamp string
note	TEXT	NULLABLE	Optional notes and remarks

7. Application Screenshots & Visual Tour

The figures below depict actual running screenshots of SpendWise captured on an Android device:

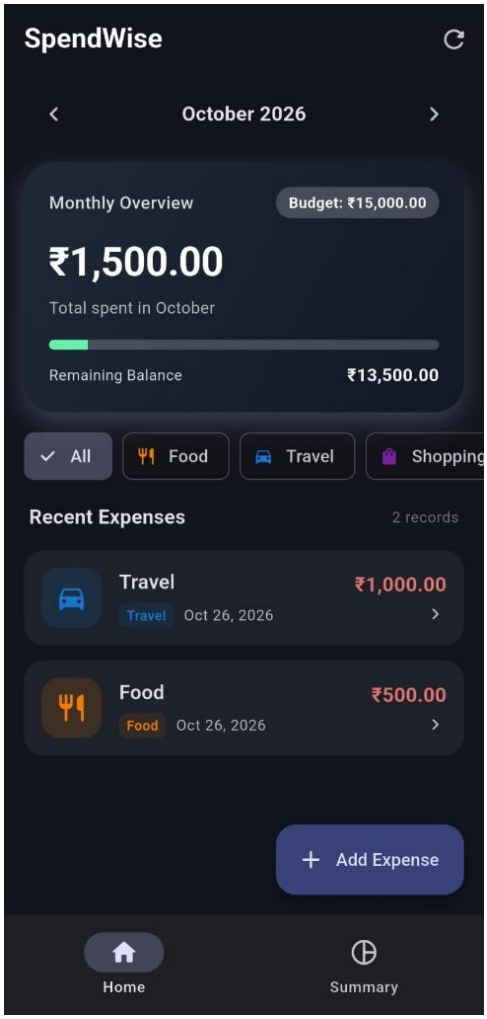


Figure 1: Home Dashboard showing monthly overview (₹1,500 spent, ₹13,500 balance), category chips, and recent expenses.

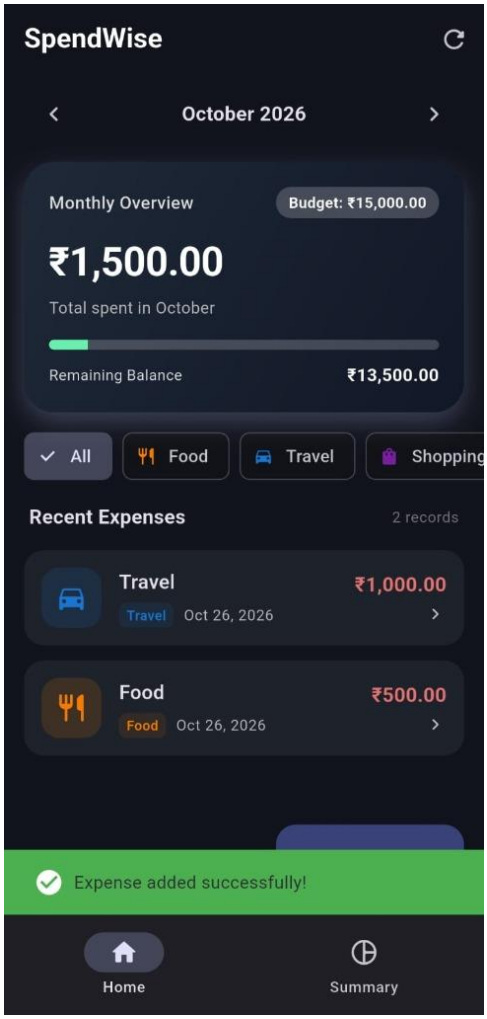


Figure 2: Instant green confirmation SnackBar displayed upon successfully adding an expense record.

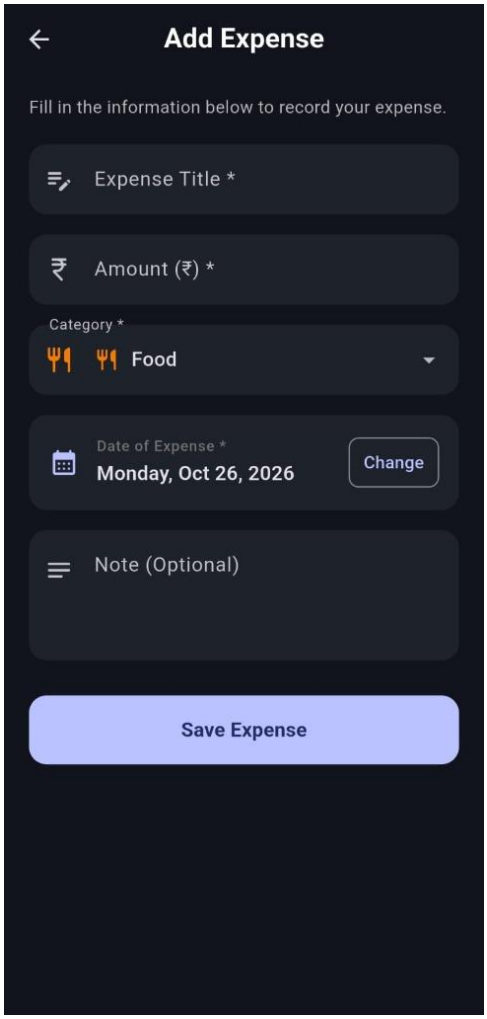


Figure 3: Add Expense Form with input validation, numeric currency field, category picker, and date selector.

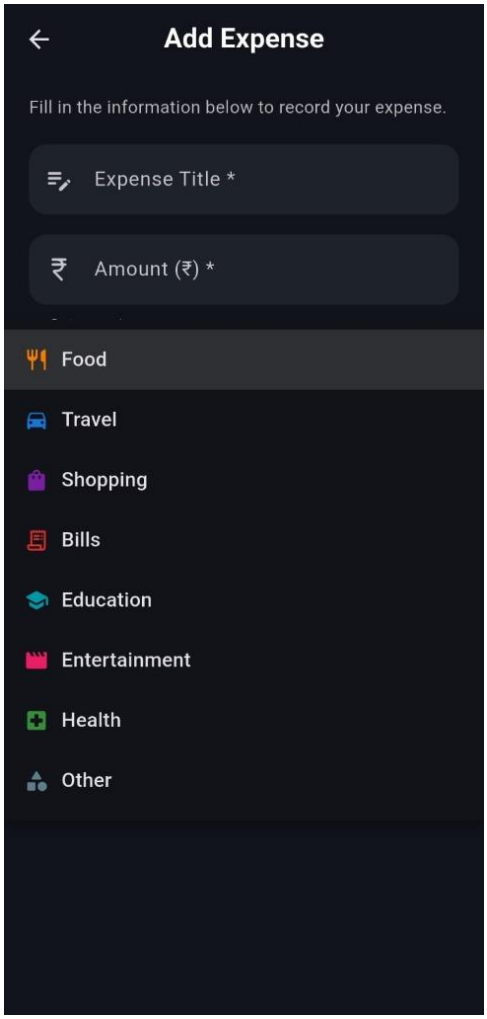


Figure 4: Category Dropdown Selector showing custom icons for Food, Travel, Shopping, Bills, Education, etc.

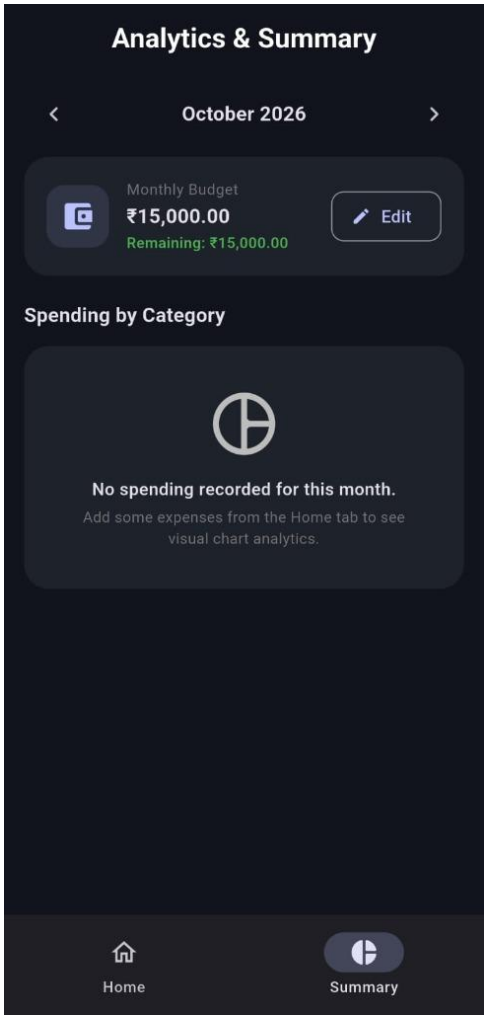


Figure 5: Analytics & Summary screen with Monthly Budget card and category expenditure breakdown.

8. Testing & Verification

10 rigorous test cases covering validation, CRUD operations, budgeting logic, and charts were executed:

TC ID	Scenario	Input Action	Expected Result	Status
TC-01	Empty Title Validation	Submit Title blank	Displays 'Title is required'	**PASSED**
TC-02	Title Min Length	Enter 1 character 'A'	Displays 'Min 2 characters required'	**PASSED**

TC-03	Non-numeric Amount	Enter 'abc' or '$50'	Displays 'Please enter valid number'	**PASSED**
TC-04	Zero/Negative Amount	Enter '0' or '-150'	Displays 'Amount must be > 0'	**PASSED**
TC-05	Expense Creation	Add 'Travel' ₹1000	Saved in SQLite; SnackBar shown	**PASSED**
TC-06	Expense Update	Edit ₹1000 to ₹1200	DB updated; total recalculated	**PASSED**
TC-07	Swipe Delete & Undo	Swipe tile, tap Undo	Record restored to SQLite list	**PASSED**
TC-08	Category Filter Chips	Tap 'Food' chip	Displays only Food expenses	**PASSED**
TC-09	Over-Budget Warning	Exceed ₹5000 budget	Red warning banner displays	**PASSED**
TC-10	Pie Chart Aggregation	View Summary tab	fl_chart displays slices matching DB	**PASSED**

9. Conclusion & Future Scope

Conclusion:

SpendWise successfully provides a production-ready, completely offline personal budgeting and expense tracking mobile application tailored for college students. By eliminating logins, remote syncs, and advertisements, it guarantees privacy and instant performance.

Future Scope:

- CSV/PDF statement export for offline financial records.
- Biometric authentication (Fingerprint / Face Unlock).
- Machine Learning receipt OCR for auto-reading paper receipts.
- Recurring automated expense schedules for monthly subscriptions.

10. References

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2. Dart Language Reference: https://dart.dev/guides
3. SQLite Architecture: https://www.sqlite.org/docs.html
4. sqflite Flutter Plugin: https://pub.dev/packages/sqflite

5. fl_chart Package: https://pub.dev/packages/fl_chart
6. GitHub Repository: https://github.com/HeelSoni/SpendWise_MAD